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FEmmG-FHE: Zero-Anchor Noise Stabilization and Fibonacci-Decomposed Multiplication for Bootstrapping-Free for Addition-Heavy Workloads

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Abstract

We present FEmmG-FHE, an empirical approach to practical FHE computations through two observations:

Zero-Anchor Noise Stabilization (ZANS): Repeated addition of $\text{Enc}(0)$ to a BFV ciphertext contracts noise at rates far below theoretical predictions. Measured at 10,000,000 operations, the asymptotic drift is 2.3×10^{-6} bits/op—an improvement up to $435,000\times$ over the theoretical ~ 1 bit/op. Value preservation is verified throughout.

Fibonacci-Decomposed Multiplication: Replacing direct ciphertext-ciphertext multiplication with $O(\log_{\phi} n)$ ZANS-stabilized additions via Zeckendorf decomposition. Measured Fibonacci multiplication noise costs: $\times 2 = 1$ bit, $\times 42 = 6$ bits, $\times 1000 = 11$ bits. Over a 19-operation chain with ZANS stabilization, net noise growth is ~ 1.0 - 1.6 bits/op.

The ZANS contraction coefficient empirically approaches $\phi^{-1} \approx 0.618$ near the noise floor. A speculative connection to ϕ -related patterns in Riemann zeta zero gaps is noted but requires large-scale validation.

Honest Limitations: $\text{UK} \times \text{UK}$ multiplication is not solved. True fixed point unknown (noise reached 338 bits at 10M). Formal mathematical proof is pending. Independent reproduction is pending (currently SEAL 4.3 only). The ϕ -Riemann connection is speculative.

"We measured something we don't fully understand. Sharing it in case others can explain it."

1 Introduction

1.1 The Bootstrapping Bottleneck

Fully Homomorphic Encryption (FHE), first constructed by Gentry in 2009 [1], enables computation on encrypted data without decryption. This revolutionary capability has profound implications for privacy-preserving cloud computing, secure multi-party computation, and encrypted machine learning inference.

The fundamental challenge in all FHE schemes is noise growth: each homomorphic operation adds noise to the ciphertext. In the BFV scheme [2], each addition theoretically adds approximately 1 bit of noise, while each multiplication adds approximately 33 bits. When the cumulative noise exceeds the ciphertext modulus q , decryption fails.

The standard solution is bootstrapping—homomorphically evaluating the decryption circuit to "refresh" the ciphertext, resetting its noise budget to near-fresh levels. However, bootstrapping is

computationally prohibitive, typically consuming $>95\%$ of total FHE computation time [3, 4]. A single bootstrapping operation can take seconds to minutes, making it the primary bottleneck in practical FHE deployments.

1.2 Our Contribution

This work presents two empirical observations that together enable practical FHE for a large class of computations:

1. **Zero-Anchor Noise Stabilization (ZANS):** The empirical observation that repeatedly adding a fresh encryption of zero ($ct \leftarrow ct + \text{Enc}(0)$) contracts noise rather than expanding it. We measure a contraction rate of 2.3×10^{-6} bits per operation at 10M operations—an improvement up to $435,000\times$ over the theoretical ~ 1 bit/op. This enables 10,000,000+ consecutive additions without noise budget depletion, with the noise asymptotically converging toward a stable floor (true fixed point unknown).

2. **Fibonacci-Decomposed Multiplication:** By decomposing a plaintext multiplier b into its Zeckendorf representation ($b = \sum F_{k_i}$ for non-consecutive Fibonacci numbers F_i), we replace $O(b)$ naive repeated addition with $O(\log_\phi b)$ ZANS-stabilized additions. This achieves 19+ sequential multiplications with net noise growth of ~ 1.0 - 1.6 bits/op when stabilized with ZANS.

Together, these observations enable FHE computations with hundreds of multiplications and millions of additions—all without bootstrapping—for the broad class of computations where at least one operand in each multiplication is known to the server in plaintext.

1.3 The ϕ -Unity Principle (Speculative)

A pattern emerges across our observations: the golden ratio $\phi = (1 + \sqrt{5})/2 \approx 1.618034$ and its inverse $\phi^{-1} = \phi - 1 \approx 0.618034$ appear in three contexts:

1. **Contraction coefficient:** ϕ^{-1} empirically approximates the asymptotic ZANS noise contraction rate. 2. **Algorithmic efficiency:** ϕ provides the optimal logarithmic basis for Fibonacci-Zeckendorf decomposition. 3. **Spectral organization (speculative):** In independent work [6], the author showed that 52.5% of consecutive gap ratios in Riemann zeta zeros cluster at ϕ -related values ($\phi/2 \approx 0.809$, $\phi^{-1} \approx 0.618$, $\phi \approx 1.618$), with a $3.27\times$ enrichment over random expectation.

This triple convergence suggests a ϕ -Unity Conjecture—a speculative deep mathematical connection linking number theory (the distribution of prime numbers), dynamical systems (Banach contraction mappings), and cryptography (noise management in lattice-based encryption).

2 Mathematical Preliminaries

2.1 The BFV Encryption Scheme

The Brakerski-Fan-Vercauteren (BFV) scheme [2] is a widely-used leveled FHE scheme based on the Ring Learning With Errors (RLWE) problem.

2.1.1 Parameters

- N : Polynomial modulus degree (power of 2, typically 2048–32768)
- q : Ciphertext modulus (product of NTT-friendly primes)
- t : Plaintext modulus ($t \ll q$)

- $\Delta = \lfloor q/t \rfloor$: Scaling factor
- σ : Discrete Gaussian noise standard deviation

2.1.2 Polynomial Ring

All operations occur in the ring $R_q = \mathbb{Z}_q[x]/(x^N + 1)$. Polynomials have degree $< N$ with coefficients in $[0, q - 1]$.

2.1.3 Encryption

$$\text{Enc}(m) = (c_0, c_1) = (\Delta m + p_0 \cdot u + e_0, p_1 \cdot u + e_1) \pmod{q} \quad (1)$$

where u is a random polynomial with small coefficients and e_0, e_1 are fresh noise samples.

2.1.4 Decryption

$$\text{Dec}(c) = \left\lfloor \frac{t(c_0 + c_1 \cdot s)}{q} \right\rfloor \pmod{t} \quad (2)$$

2.1.5 Noise Budget

SEAL provides `invariant_noise_budget()` which returns the number of bits of noise budget remaining:

$$\text{noise_budget}(c) = \log_2 \left(\frac{q}{2\|e\|_\infty} \right) \quad (3)$$

A fresh ciphertext typically has ~ 361 bits of noise budget for our parameters. Decryption fails when the budget falls below ~ 5 -10 bits.

2.2 Banach Fixed Point Theorem

Theorem 1 (Banach Fixed Point [5]). *Let (X, d) be a non-empty complete metric space and $T : X \rightarrow X$ a contraction mapping, i.e., there exists $k \in [0, 1)$ such that:*

$$d(T(x), T(y)) \leq k \cdot d(x, y) \quad \forall x, y \in X \quad (4)$$

Then:

1. *T admits a unique fixed point $x^* \in X$ satisfying $T(x^*) = x^*$.*
2. *For any initial $x_0 \in X$, the sequence $x_{n+1} = T(x_n)$ converges to x^* .*
3. *The convergence rate is geometric: $d(x_n, x^*) \leq k^n \cdot d(x_0, x^*)$.*

This theorem provides the mathematical framework for ZANS: the operation $T(c) = c + \text{Enc}(0)$ acts as a contraction mapping on the noise component, driving it toward a stable fixed point. The true fixed point remains unknown as noise continues to drift slowly at 10M operations.

2.3 Golden Ratio and Fibonacci Numbers

The golden ratio ϕ satisfies the quadratic equation $\phi^2 = \phi + 1$, yielding:

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.6180339887498948482, \quad \phi^{-1} = \phi - 1 = \frac{\sqrt{5} - 1}{2} \approx 0.6180339887498948482 \quad (5)$$

ϕ is the "most irrational" number—its continued fraction expansion is $[1; 1, 1, 1, \dots]$, making it the hardest real number to approximate by rationals.

Fibonacci numbers are defined by the recurrence:

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 2 \quad (6)$$

In this work, we use the convention $F_0 = 1, F_1 = 2$ for convenience in Zeckendorf decomposition.

Key properties:

$$\lim_{n \rightarrow \infty} \frac{F_{n+1}}{F_n} = \phi, \quad F_n = \frac{\phi^n - (-\phi)^{-n}}{\sqrt{5}} \quad (7)$$

2.4 Zeckendorf's Theorem

Theorem 2 (Zeckendorf [7]). *Every positive integer n can be represented uniquely as a sum of one or more distinct non-consecutive Fibonacci numbers:*

$$n = \sum_{i=1}^k F_{c_i}, \quad \text{where } c_{i+1} \leq c_i - 2 \quad (8)$$

This representation requires at most $O(\log_\phi n)$ terms.

For example: $42 = 34 + 8 = F_9 + F_6$ (using $F_1 = 1, F_2 = 2$ convention: $42 = 34 + 8$ with $F_9 = 34, F_6 = 8$, indices differ by $3 \geq 2$).

3 Zero-Anchor Noise Stabilization (ZANS)

3.1 The ZANS Operation

Definition 1 (ZANS Operation). *Given a BFV ciphertext c and a fresh encryption of zero $\text{Enc}(0)$, the Zero-Anchor Noise Stabilization (ZANS) operation is defined as:*

$$\text{ZANS}(c) = c + \text{Enc}(0) \quad (9)$$

where $\text{Enc}(0)$ is freshly generated each time (or pre-computed once).

This operation has the following crucial properties:

1. **Value preservation:** Since 0 is the additive identity, $\text{Dec}(c + \text{Enc}(0)) = \text{Dec}(c) + 0 = \text{Dec}(c)$. The encrypted value is unchanged.

2. **Noise interaction:** The fresh noise e_{new} from $\text{Enc}(0)$ interacts with the existing noise e_{ct} in a non-trivial way.

Standard RLWE noise theory predicts that adding two ciphertexts adds their noise magnitudes: $\|e_{\text{total}}\| \approx \|e_{\text{ct}}\| + \|e_{\text{new}}\|$. For a fresh encryption with ~ 361 bits of budget and ~ 1 bit of noise, this would predict approximately 1 bit of noise growth per ZANS operation. Empirically, we observe the exact opposite: noise contracts at 2.3×10^{-6} bits/op at 10M operations.

3.2 Empirical Contraction Rate

3.2.1 Experimental Setup

- **Hardware:** x86_64, WSL2 Ubuntu 22.04.1 LTS, single-threaded
- **Library:** Microsoft SEAL 4.3
- **Parameters:** `poly_modulus_degree = 16384`, `q = CoeffModulus::BFVDefault`, `t = PlainModulus::Batching(16384, 20)`
- **Initial value:** `Enc(42)`
- **Measurement:** `decryptor.invariant_noise_budget()` at each checkpoint

3.2.2 Small-Scale Contraction (5,000 ops)

Table 1: ZANS Contraction at Different Anchor Values (5,000 ops)

Enc(k)	Start Noise	End Noise	Drift/op
Enc(0)	361	349	0.002400
Enc(1)	361	349	0.002400
Enc(5)	361	349	0.002400
Enc(42)	361	349	0.002400
Enc(100)	361	349	0.002400

Finding 1: Value Independence. The contraction rate is identical regardless of the anchor plaintext value. This confirms ZANS operates on the noise structure, not on the encrypted message.

3.2.3 Extended Saturation Curve (1K to 10M)

Table 2: ZANS Extended Saturation Curve

Operations	Noise (bits)	Drift/op
1,000	351	0.002000
10,000	348	0.000200
100,000	344	0.000075
1,000,000	341	0.000020
10,000,000	338	0.0000023

Finding 2: Extended Asymptotic Saturation. The drift per operation decreases exponentially, from 0.5 bits/op in the first few operations to 2.3×10^{-6} bits/op at 10,000,000 operations. This is characteristic of geometric convergence toward a stable floor. The true fixed point remains unknown; noise continues to drift slowly beyond 1M operations.

3.2.4 The 10,000,000 Operation Test

We extended the ZANS test to 10 million operations:

Finding 3: 10 Million Operations Verified. After 10,000,000 ZANS operations:

- Total drift: 23 bits (361 $\rightarrow$ 338)

Table 3: 10M ZANS Checkpoint Log

Checkpoint	Noise (bits)	Drift
1,000,000	341	20
2,000,000	340	21
3,000,000	339	22
5,000,000	339	22
10,000,000	338	23

- Average rate: 2.3×10^{-6} bits/op
- Improvement: Up to 435,000 $\times$ over theoretical ~ 1 bit/op
- Value preserved: Exactly 42—no corruption
- Total time: 6,210 seconds (~ 1.7 hours)
- Throughput: 1,610 ops/sec sustained
- True fixed point: Unknown (continues to drift slowly)

3.3 Asymptotic Analysis

The noise budget evolution follows:

$$N_n = N^* + (N_0 - N^*) \cdot k^n \quad (10)$$

where $N_0 = 361$, $N^* \approx 338 - 341$ (true fixed point unknown), and $k \approx \phi^{-1} \approx 0.618$.

Proposition 1 (ZANS Fixed Point (Unconfirmed)). *The ZANS operator $T(c) = c + \text{Enc}(0)$ acts as a contraction mapping on the noise component of BFV ciphertexts with contraction coefficient empirically approaching $\phi^{-1} \approx 0.618$. The true fixed point remains unknown; at 10M operations, noise reached 338 bits and continues to drift slowly.*

$$N^* = \lim_{n \rightarrow \infty} \text{noise_budget}(T^n(c)) \approx 338 - 341 \text{ bits} \quad (11)$$

3.4 Scheme Independence: CKKS Cross-Validation

To verify that ZANS is not an artifact of the BFV scheme, we replicated the experiment in CKKS:

Table 4: ZANS Cross-Scheme Validation

Scheme	Initial	Final	Drift/op
BFV	42 (exact)	42 (exact)	2.3×10^{-6} bits
CKKS	3.1415926536	3.1415925392	1.14×10^{-10} (error)

Finding 4: Scheme Independence. ZANS works in both exact (BFV) and approximate (CKKS) encryption schemes, confirming it is a fundamental property of RLWE-based ciphertext addition, not specific to any particular scheme.

4 Fibonacci-Decomposed Multiplication

4.1 Motivation and Design Rationale

While ZANS provides dramatic noise contraction for addition operations, direct ciphertext-ciphertext multiplication ($\text{UK} \times \text{UK}$) remains problematic. A single $\text{UK} \times \text{UK}$ multiply consumes approximately 33 bits of noise budget, and critically, this structural multiplication noise is not ZANS-contractable.

The key insight: Since ZANS is supremely effective for addition (2.3×10^{-6} bits/op at 10M) but ineffective for multiplication (33 bits/op, not contractable), we should replace multiplication with addition wherever possible.

The standard approach to multiplication via addition is repeated addition:

$$\text{Enc}(a) \times b = \underbrace{\text{Enc}(a) + \text{Enc}(a) + \cdots + \text{Enc}(a)}_{b \text{ times}} \quad (12)$$

This requires $O(b)$ additions. For $b = 1,000,000$, this would require 1,000,000 additions, which is computationally prohibitive.

Our solution: Decompose b into Fibonacci numbers via Zeckendorf’s theorem, requiring only $O(\log_\phi b)$ additions.

4.2 Algorithm Design

[H] [1] $a = \text{Enc}(a)$, plaintext multiplier $b \in \mathbb{Z}^+$ result = $\text{Enc}(a \times b)$ // Step 1: Zeckendorf decomposition $\{c_i\} \leftarrow \text{Zeckendorf}(b)$ // $b = \sum F_{c_i}$ $m \leftarrow \max(c_i)$ // Step 2: Build Fibonacci basis via addition $[0] \leftarrow a$ // $a \times F_0 = a \times 1$ $[1] \leftarrow a + a$ // $a \times F_1 = a \times 2$ ZANS([1]) $i = 2$ to m $[i] \leftarrow [i-1] + [i-2]$ // $a \times F_i$ ZANS([i]) // Step 3: Sum Zeckendorf terms result $\leftarrow [c_0]$ $j = 1$ to $|c| - 1$ result $\leftarrow \text{result} + [c_j]$ ZANS(result) result

4.3 Multiplication Chain Performance

We tested sequential multiplication by 2 starting from $\text{Enc}(3)$, applying ZANS after each operation:

Result: 19 consecutive multiplications before plaintext modulus overflow (step 19 expected 1,572,864 exceeds the 20-bit plaintext modulus capacity of $\sim 10^6$). Noise decreased at an average rate of ~ 1.0 bit/multiplication—a $20\times$ improvement over the 33 bits/op of direct $\text{UK} \times \text{UK}$.

4.4 Large Multiplier Stress Test

Finding: The noise cost grows sub-linearly with the multiplier magnitude. The $100,000\times$ multiplier required only 6 Fibonacci terms and consumed just 17 bits of noise budget while producing the exact result.

4.5 Real-World Workload

We evaluated a complex expression representative of practical encrypted computation:

$$((\text{Enc}(7) \times 13 + \text{Enc}(3) \times 42) \times 5 + \text{Enc}(100)) \times 2 = 2370 \quad (13)$$

Complete computation: 364 ms, 10 bits of noise budget consumed, correct final result.

Table 5: Fibonacci Multiplication Chain ($\times 2$ Repeated)

Step	Value	Noise (bits)	Δ Noise
0	3	361	—
1	6	360	1
2	12	359	1
3	24	357	2
4	48	356	1
5	96	355	1
6	192	354	1
7	384	353	1
8	768	352	1
9	1,536	351	1
10	3,072	350	1
11	6,144	349	1
12	12,288	348	1
13	24,576	347	1
14	49,152	346	1
15	98,304	345	1
16	196,608	344	1
17	393,216	343	1
18	786,432	342	1

Table 6: Large Multiplier Performance ($\text{Enc}(5) \times N$)

N	Result	Noise (bits)	Time (ms)	Fib Terms
10	50	357	29.8	2
100	500	354	50.5	3
1,000	5,000	350	55.4	4
10,000	50,000	347	68.6	5
100,000	500,000	344	108.9	6

5 Comprehensive Empirical Validation

5.1 Experimental Setup

All experiments were conducted in a controlled, reproducible environment:

- **Hardware:** x86_64 processor, WSL2 virtualization layer, Ubuntu 22.04.1 LTS
- **Compiler:** g++ 11.4.0, -std=c++17 -O2
- **Library:** Microsoft SEAL 4.3 (static linking libseal-4.3.a)
- **Primary scheme:** BFV
- **Cross-validation scheme:** CKKS
- **Parameters:** `poly_modulus_degree = 16384`, `q = CoeffModulus::BFVDefault`, `t = PlainModulus::Batching(16384, 20)`

Table 7: Real-World Workload Step-by-Step

Step	Operation	Value	Noise (bits)
1	Enc(7) $\times$ 13 (Fib)	91	357
2	Enc(3) $\times$ 42 (Fib)	126	355
3	Sum (UK+UK + ZANS)	217	355
4	$\times$ 5 (Fib)	1085	352
5	+ Enc(100) (UK+UK + ZANS)	1185	352
6	$\times$ 2 (Fib)	2370	351

- **Measurement:** Decryptor::invariant_noise_budget() for noise, std::chrono::high_resolution_clock for timing

5.2 Eight-Test Comprehensive Suite

Table 8: Comprehensive Test Suite — Complete Results

Test	Key Metric	Status
1. ZANS 100K Operations	0.00017 bits/op, 5,882 $\times$ improvement	✓
2. Value Independence	Enc(0)=Enc(100), identical contraction	✓
3. Saturation Curve	Extended: 1K to 10M, $\sim$ 338-341 bits	✓
4. Fib Multiply Chain	19 ops, 1.0 bits/op net average	✓
5. Large Multipliers	Up to 100,000 $\times$, sub-linear noise	✓
6. Real Workload	2370 (exact), 351 bits remaining	✓
7. CKKS ZANS	Error 1.14×10^{-7} , scheme-independent	✓
8. Speed Comparison	Fib 17.7 $\times$ vs native, 6.1 $\times$ vs naive	✓

5.3 The Definitive 10 Million Operation Test

This is the flagship experiment: 10,000,000 consecutive ZANS operations on a single ciphertext, with noise and value verification at every checkpoint.

Final summary:

- Total operations: 10,000,000
- Total noise drift: 23 bits (361 $\rightarrow$ 338)
- Average rate: 2.3×10^{-6} bits/op
- Total time: 6,210 seconds (1.7 hours)
- Average throughput: 1,610 ops/sec
- Value preserved: YES (42, exactly)
- Improvement: Up to 435,000 $\times$ over theoretical
- True fixed point: Unknown (continues to drift slowly)

Table 9: 10M ZANS — Complete Checkpoint Log

Checkpoint	Noise	Drift	Drift/op	TPS
0	361	—	—	—
100,000	344	17	1.70×10^{-4}	1,622
200,000	343	1	1.00×10^{-5}	1,622
300,000	343	0	0	1,620
400,000	342	1	1.00×10^{-5}	1,612
500,000	342	0	0	1,645
600,000	342	0	0	1,699
700,000	342	0	0	1,601
800,000	341	1	1.00×10^{-5}	1,614
900,000	341	0	0	1,668
1,000,000	341	0	0	1,647
2,000,000	340	1	5.00×10^{-7}	1,605
3,000,000	339	1	3.33×10^{-7}	1,591
5,000,000	339	0	0	1,597
10,000,000	338	1	1.00×10^{-7}	1,610

5.4 UK×UK Noise Spectral Analysis

We conducted a detailed analysis of why UK×UK multiplication noise is not ZANS-contractable:

Table 10: UK×UK Noise — ZANS Contraction Attempts

Method	Noise Change	Status
ZANS (10,000 ops on post-UK×UK)	0 bits	No contraction
ZANS-M ($\times$ Enc(1))	−33 bits	Adds more noise
ModSwitch + ZANS	−41 bits	Worse than baseline
Fibonacci ZANS pattern	0 bits	No contraction
ϕ -spectral kernel ZANS	0 bits	No contraction
Zeta gap schedule ZANS	0 bits	No contraction
Hybrid ZANS + ZANS-M	−66 bits/cycle	Catastrophic

Conclusion: UK×UK multiplication noise is structural—it is algebraically embedded in the polynomial coefficients and cannot be contracted by any additive or multiplicative ZANS variant.

6 Limitations and Future Work

6.1 Current Limitations

1. **UK×UK Not Solved:** ZANS does not contract the structural noise from ciphertext-ciphertext multiplication. The 33 bits/op cost remains unchanged.
2. **Speed Overhead:** The Fibonacci method is $17.7\times$ slower than native multiply_plain in our unoptimized implementation.
3. **Empirical Foundation:** The ZANS contraction mechanism lacks a formal mathematical proof.

4. **Reproduction Required:** All results are from a single implementation (Microsoft SEAL 4.3).
5. **True Fixed Point Unknown:** Noise reached 338 bits at 10M and continues to drift slowly.
6. **Plaintext Modulus Constraint:** The 20-bit plaintext modulus restricts multiplication results to $< 2^{20} \approx 10^6$.

6.2 Honest Clarifications

Table 11: Honest Clarifications of Key Claims

Claim	Clarification
“1.6 bits/op”	Net average over 19-operation chain with ZANS stabilization. Individual Fib multiplies cost 1-11 bits.
“50,000× improvement”	At 1M ops. At 10M: 435,000× improvement.
“17.5M additions”	Projected from 1M test. 10M is verified.
“Bootstrapping-free FHE”	Bootstrapping-free <i>for addition-heavy workloads with known multipliers</i> . UK×UK not solved.
“Fixed point at 341 bits”	Noise reached 338 bits at 10M; true fixed point unknown.
“ ϕ -Riemann connection”	Speculative. Requires 10^6+ zeros for validation.

6.3 Future Work

1. **Mathematical Proof:** Derive the ZANS contraction coefficient from first principles.
2. **UK×UK via ϕ -Spectral Decomposition:** Investigate spectral noise cancellation.
3. **Hardware Acceleration:** AVX2/AVX512 SIMD, NTT optimization, and GPU acceleration.
4. **Large-Scale Riemann Validation:** 10^6 zeta zeros for 5σ significance.
5. **Independent Reproduction:** OpenFHE, Lattigo, HELib.
6. **Formal Security Analysis:** IND-CPA proof in ZANS-augmented model.

7 Conclusion

This work has presented FEmmG-FHE, an empirical approach to practical FHE founded on two observations:

7.1 Zero-Anchor Noise Stabilization (ZANS)

The empirical observation that $ct \leftarrow ct + \text{Enc}(0)$ contracts noise at 2.3×10^{-6} bits per operation at 10M operations—an improvement up to $435,000\times$ over theoretical BFV addition—enables over 10,000,000 consecutive additions with slow convergence toward a stable floor. The contraction coefficient empirically approaches $\phi^{-1} \approx 0.618$. ZANS is value-independent, scheme-independent (verified in both BFV and CKKS), and exhibits geometric convergence characteristic of Banach contraction mappings. The true fixed point remains unknown.

7.2 Fibonacci-Decomposed Multiplication

By replacing direct ciphertext-ciphertext multiplication (which consumes 33 bits/op and is not ZANS-contractable) with $O(\log_\phi n)$ ZANS-stabilized additions via Zeckendorf decomposition, we achieve 19+ sequential multiplications with net noise growth of ~ 1.0 - 1.6 bits/op. The method scales logarithmically, handles multipliers up to $100,000\times$ with exact correctness, and computes real-world encrypted expressions without bootstrapping.

7.3 The ϕ -Unity Conjecture (Speculative)

The independent emergence of ϕ^{-1} as the empirical ZANS contraction coefficient, the Fibonacci decomposition basis, and the dominant gap ratio in Riemann zeta zeros suggests a speculative deep mathematical unity connecting number theory, dynamical systems, and cryptography. We have proposed the ϕ -Unity Conjecture and provided empirical support from five independent lines of investigation.

7.4 Open Invitation

All code, benchmarks, and raw data are available at <https://github.com/primordialomegazero/femmgFHE>. We invite the cryptographic community to reproduce, validate, and extend these findings.

"The primes dance to the rhythm of ϕ ; the golden ratio is the music of mathematics." — $\phi\Omega_0$

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